Camunda DMN Decision Table HIT policy Any

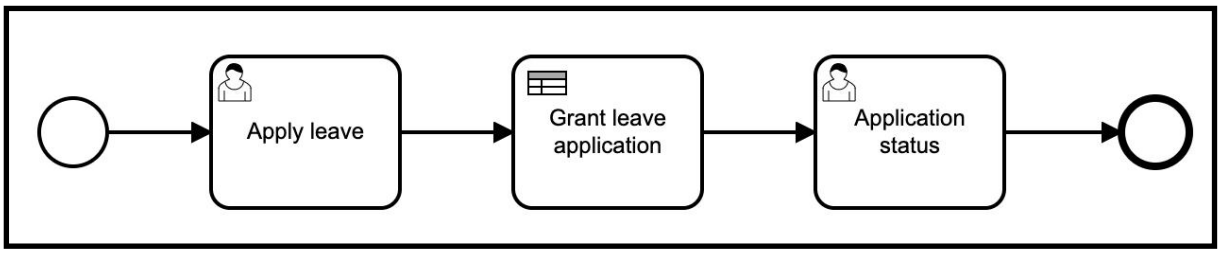
In this session we will try to understand what is decision table with HIT policy ANY

With the HIT policy ANY, multiple rules can be satisfied. However, all satisfied rules must generate the same output. The decision table result contains only the output of one of the satisfied rules.

If multiple rules are satisfied which generate different outputs, the hit policy is violated.

To demonstrate the user of HIT policy ANY, We will model below process and decision table.

1. Process definition (BPMN)



2. Decision table (DMN)

Edit DRD | Open Overview

Grant leave | Hit Policy: Any

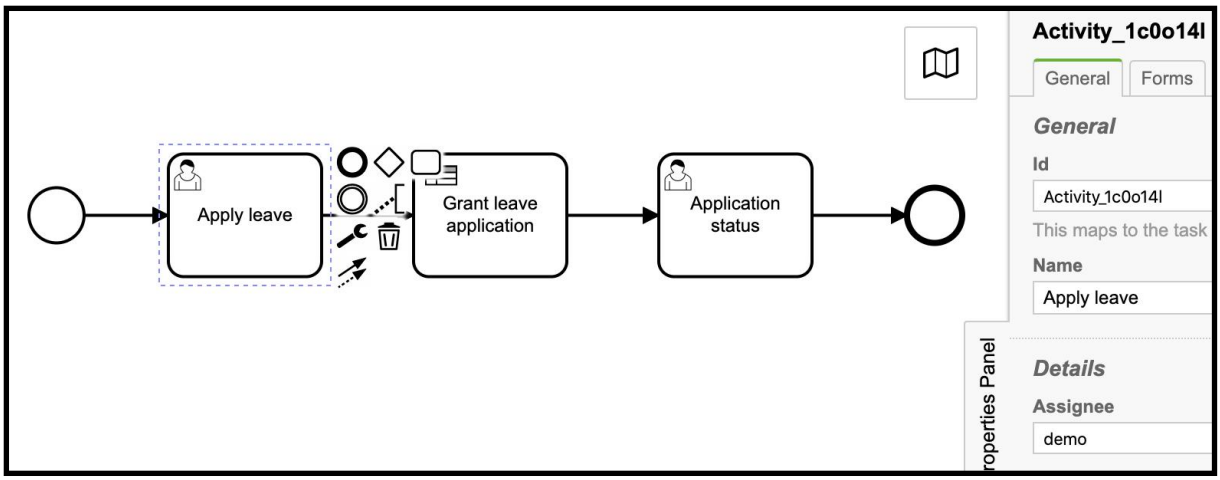
	When Vacation days integer	And Status string	Then Application status string
1	0	-	"Rejected"
2	> 0	"On probation period"	"Rejected"
3	> 0	"Not on probarion period"	"Approved"
+	-	-	

3. Process definition properties

Id: Process_HIT_policy_any

Name: Process HIT Policy Any

3.1. User task: Guest Registration

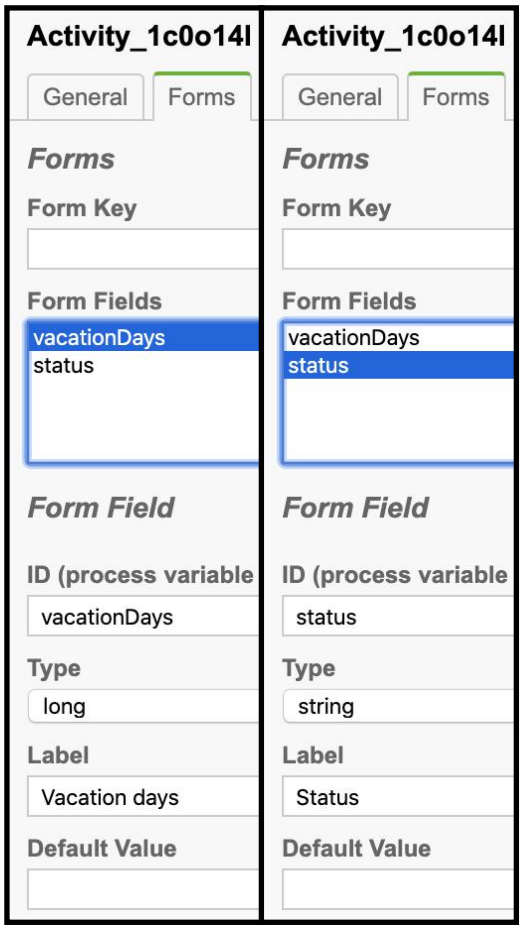


Id: Activity_1c0o14l

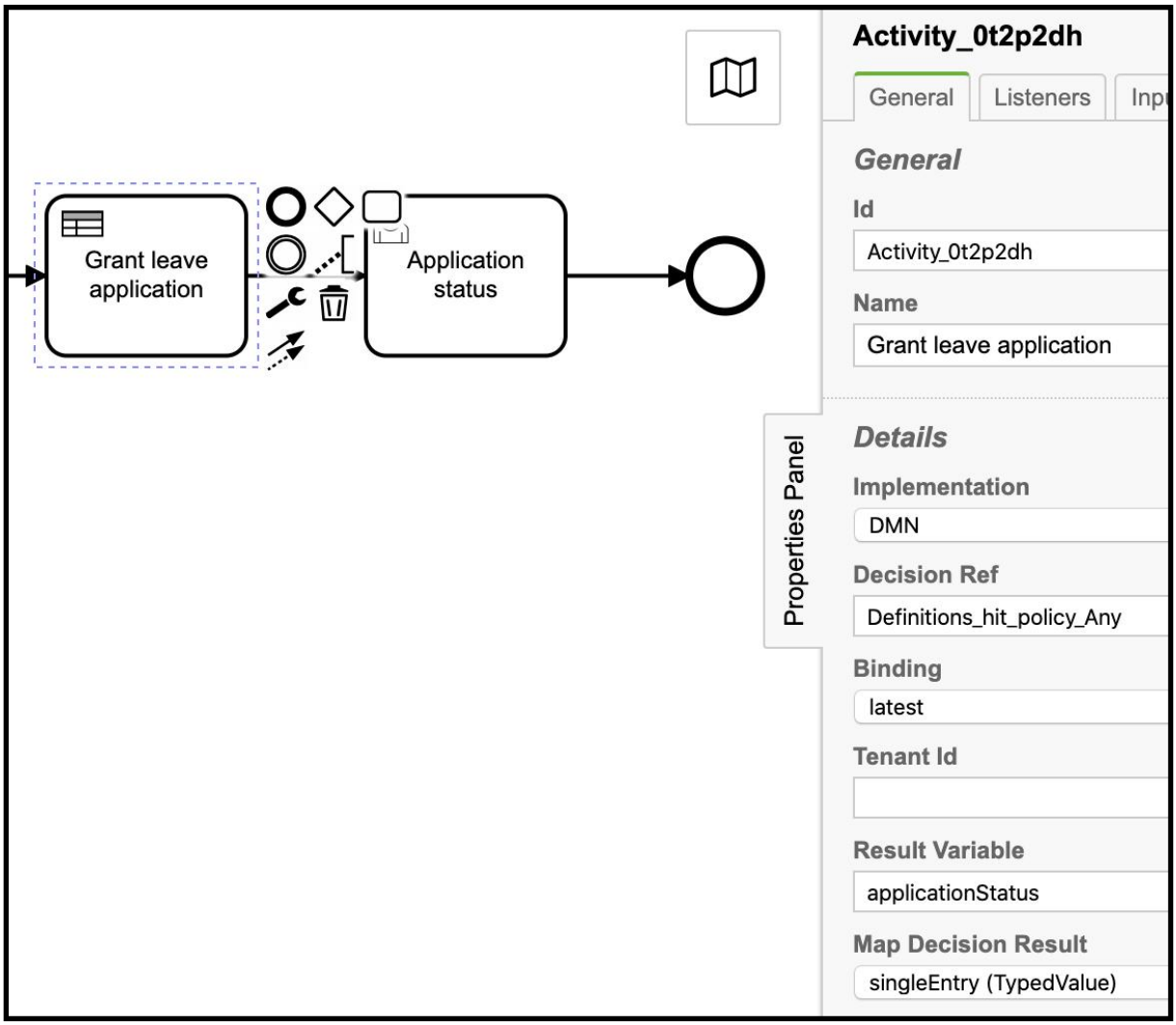
Name: Apply leave

Assignee: demo

Task form



3.2. Business rule task: Decide Leave



Id: Activity_0t2p2dh

Name: Grant leave application

Implementation: DMN

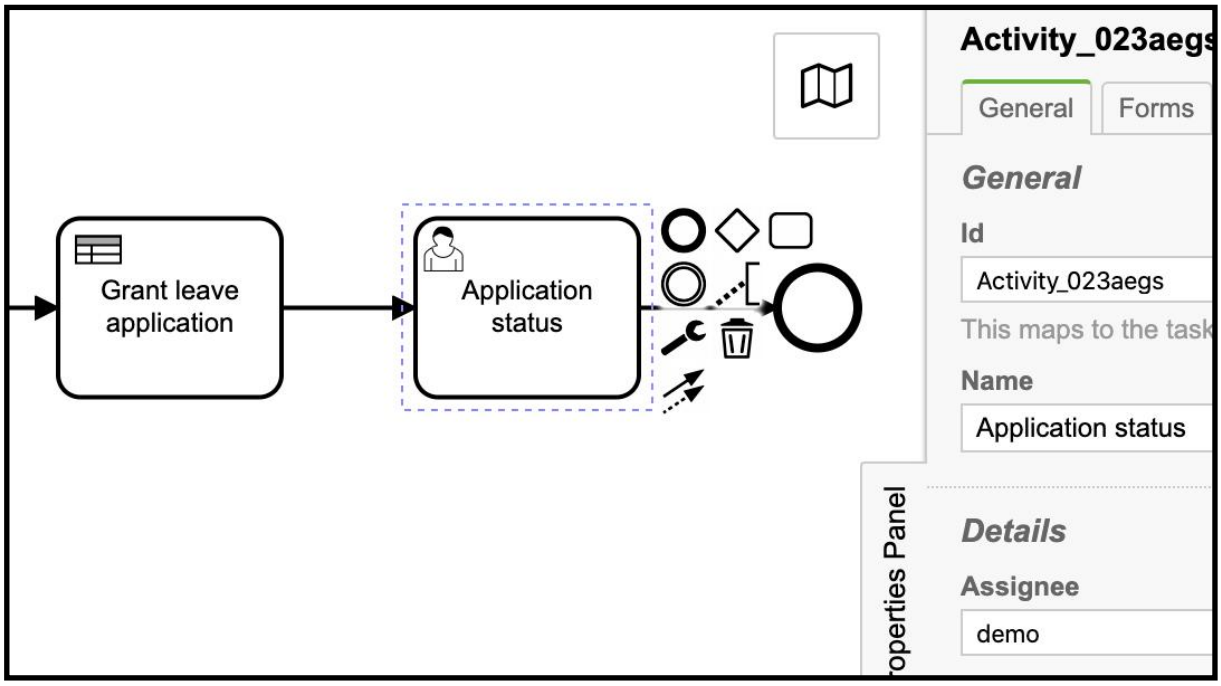
Decision Ref: Definitions_hit_policy_Any

Binding: latest

Result Variable: applicationStatus

Map Decision Result: singleEntry (TypedValue)

3.3. User task: Application status

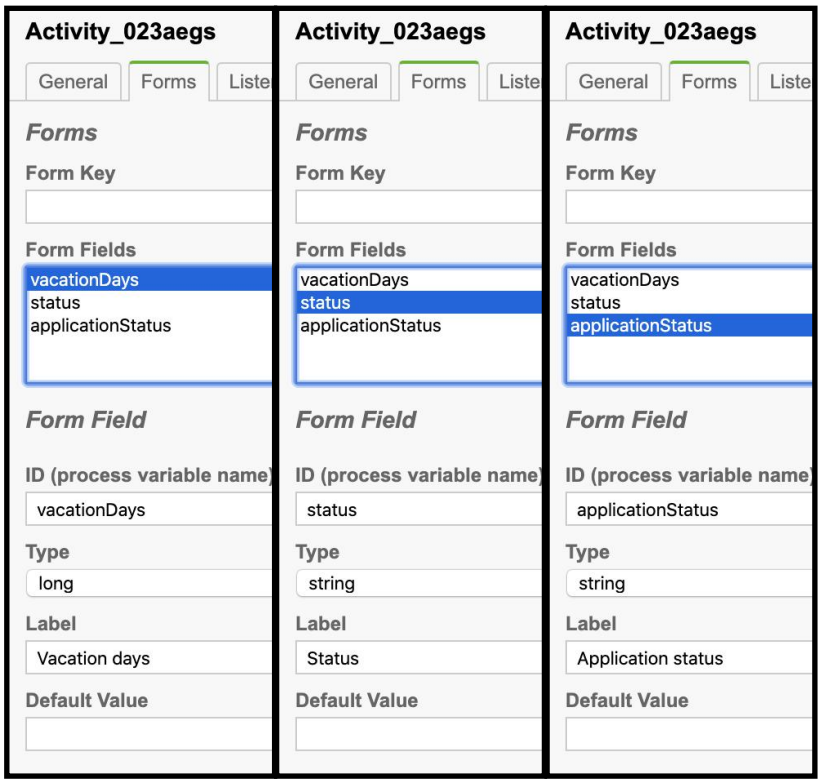


Id: Activity_023aegs

Name: Application status

Assignee: demo

Task form



4. Decision table:

Edit DRD | Open Overview

Grant leave | Hit Policy: Any

	When Vacation days integer	And Status string	Then Application status string
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+	-	-	

Properties:

Definitions_hit_policy_Any

General

General

Id

Definitions_hit_policy_Any

This maps to the decision definition key.

Name

Grant leave

Version Tag

History Configuration

History Time To Live

Id: Definitions_hit_policy_Any

Name: Grant leave

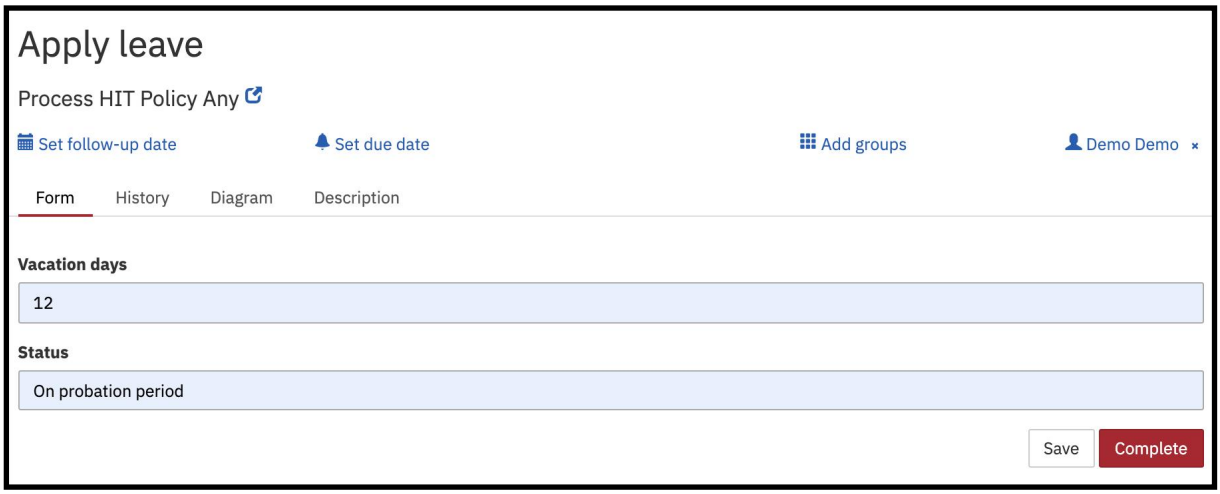
5. Deploy process definition and decision table.

6. Testing the process definition and DMN

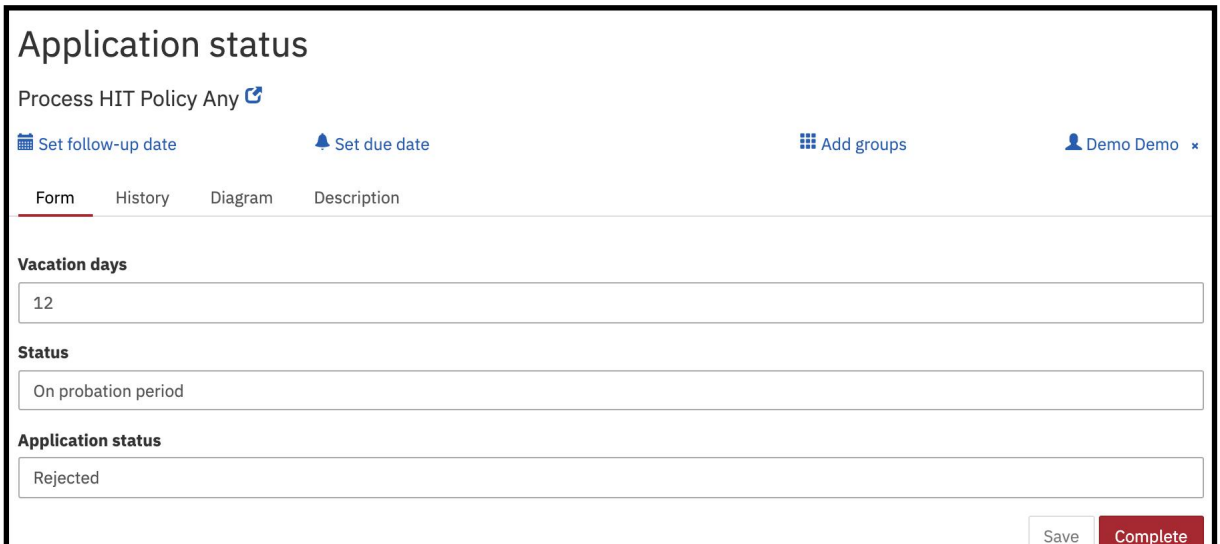
Start the process definition

Hit **Process HIT Policy Any** and press Start button to start process.

As per process definition, user task Apply leave will get created.



Fill-up the task form and press Complete button and then as per process definition business rule task get executed and will produce single output based on provided data by task and same output result can be viewed on newly created user task Application status.



It can be seen that business rule task has invoked decision table and produces output applicationStatus = Rejected

Note: In decision table if you have two rules satisfied with different output by provided input data then error will be thrown as below

An error happened while submitting the task form :
DMN-03002 Hit policy 'ANY' only allows multiple matching rules with equal output. Matching rules: '[DmnEvaluatedDecisionRuleImpl{id='DecisionRule_0j3lozd', outputEntries={applicationStatus=DmnEvaluatedOutputImpl{id='Output_1', name='Application status', outputName='applicationStatus', value=Value 'Rejected' of type 'PrimitiveValueType[string]', isTransient=false}}}, DmnEvaluatedDecisionRuleImpl{id='DecisionRule_05glf6u', outputEntries={applicationStatus=DmnEvaluatedOutputImpl{id='Output_1', name='Application status', outputName='applicationStatus', value=Value 'Approved' of type 'PrimitiveValueType[string]', isTransient=false}}}]'.

Tomcat console error logs

14-Sep-2020 23:48:20.359 WARNING [http-nio-8080-exec-10] org.camunda.commons.logging.BaseLogger.logWarn ENGINE-REST-HTTP500 org.camunda.bpm.dmn.engine.impl.hitpolicy.DmnHitPolicyException: DMN-03002 Hit policy 'ANY' only allows multiple matching rules with equal output. Matching rules: '[DmnEvaluatedDecisionRuleImpl{id='DecisionRule_0j3lozd', outputEntries={applicationStatus=DmnEvaluatedOutputImpl{id='Output_1', name='Application status', outputName='applicationStatus', value=Value 'Rejected' of type 'PrimitiveValueType[string]', isTransient=false}}}, DmnEvaluatedDecisionRuleImpl{id='DecisionRule_05glf6u', outputEntries={applicationStatus=DmnEvaluatedOutputImpl{id='Output_1', name='Application status', outputName='applicationStatus', value=Value 'Approved' of type 'PrimitiveValueType[string]', isTransient=false}}}]'.